homework05

P185-习题 9.1 (A 类) 7-10, 7-13; 9; 11-1; 12;
17-3; 18-1; 18-2;

7-10、计算 $\iint_D \sqrt{x^2 - xy}\, d\sigma$, 其中 D 是由直线 $x = 1, y = 0, x = y$ 所围成的闭区域.

解: 依题意 $D : 0 \le x \le 1, 0 \le y \le x$

$$(*) = \int_0^1 dx \int_0^x dy (x^2 - xy)^{1/2} = \int_0^1 dx (x^2 - xy)^{3/2} \frac{-2}{3x}\Big|_0^x$$
$$= \frac{2}{3}\int_0^1 x^2\, dx = \frac{2}{9}x^3\Big|_0^1 = \frac{2}{9}$$

7-13、计算 $\iint_D \sin\frac{x}{y}\, d\sigma$, 其中 D 是由直线 $y = x, y = 2$ 与曲线 $x = y^3$ 所围的闭区域.

解: 依题意, $D : 1 \le y \le 2, y \le x \le y^3$

$$I = \int_1^2 dy \int_y^{y^3} dx \sin\frac{x}{y} = \int_1^2 dy(-y)\cos\frac{x}{y}\Big|_y^{y^3} = \int_1^2 dy\left(y\cos 1 - y\cos y^2\right)$$
$$= \frac{1}{2}y^2\cos 1 - \frac{1}{2}\sin y^2\Big|_1^2 = \frac{3}{2}\cos 1 - \frac{1}{2}\sin 4 + \frac{1}{2}\sin 1$$

9、计算抛物柱面 $z=4-x^2$, 坐标面和平面 $\pi: 2x+y=4$ 所围立体在第一卦限的部分的体积.

解: 依题意 $D: x\geq 0, y\geq 0, 2x+y\leq 4 \Rightarrow 0\leq x\leq 2, 0\leq y\leq 4-2x$

$$V=\int_0^2 dx\int_0^{4-2x} dy(4-x^2)=\int_0^2 dx\,(4-2x)(4-x^2)$$

$$=\int_0^2 \left(16-4x^2-8x+2x^3\right)dx=16x-\frac{4}{3}x^3-4x^2+\frac{x^4}{2}\Big|_0^2$$

$$=32-\frac{32}{3}-16+8=\frac{40}{3}$$

11-1、积分换序 $I=\int_0^4 dy\int_{-\sqrt{4y}}^{\sqrt{4y-y^2}} f(x,y)\,dx$

解: 依题意, $D=D_1\cup D_2$

$$D_1: 0\leq y\leq 4, -\sqrt{4y}\leq x\leq 0 \Rightarrow -4\leq x\leq 0, x^2/4\leq y\leq 4$$

$$D_2: 0\leq y\leq 4, 0\leq x\leq \sqrt{4y-y^2} \Rightarrow 0\leq x\leq 2, 2-\sqrt{4-x^2}\leq y\leq 2+\sqrt{4-x^2}$$

$$\Rightarrow I=\int_{-4}^0 dx\int_{x^2/4}^4 dy\, f(x,y)+\int_0^2 dx\int_{2-\sqrt{4-x^2}}^{2+\sqrt{4-x^2}} dy\, f(x,y)$$

12、将二重积分 $\iint_D f(x,y)\,d\sigma$ 化为不同顺序的累次积分，其中 D 是由抛物线 $y=\frac{x^2}{4}-1$ 与直线 $y=2-x$ 所围成的区域.

解：依题意

$$由 y=\frac{x^2}{4}-1, y=2-x \Rightarrow (x,y)=(2,0),(-6,8)$$

$$\Rightarrow (*)=\int_{-6}^{2} dx \int_{\frac{x^2}{4}-1}^{2-x} dy\, f(x,y)$$

$$=\int_{0}^{8} dy \int_{-2\sqrt{1+y}}^{2-y} dx\, f(x,y)+\int_{-1}^{0} dy \int_{-2\sqrt{y+1}}^{2\sqrt{y+1}} f(x,y)\,dx$$

17-3、化积分为极坐标并计算: $\int_0^1 dx \int_{1-x}^{\sqrt{1-x^2}} \sqrt{(x^2+y^2)^{-3}}\, dy$

解: 依题意, $D: 0 \le x \le 1, 1-x \le y \le \sqrt{1-x^2}$

$$\Rightarrow 0 \le r\cos\theta \le 1, 1 - r\cos\theta \le r\sin\theta \le \sqrt{1 - r^2\cos^2\theta}$$

$$\Rightarrow \theta \in [0, \frac{\pi}{2}] \Rightarrow r \le \frac{1}{\cos\theta}, r \ge \frac{1}{\sin\theta + \cos\theta}, r \le 1 \Rightarrow \frac{1}{\sin\theta + \cos\theta} \le r \le 1, \theta \in [0, \frac{\pi}{2}]$$

$$\Rightarrow (*) = \int_0^{\pi/2} d\theta \int_{\frac{1}{\sin\theta+\cos\theta}}^1 dr r^{-3} r = \int_0^{\pi/2} d\theta \frac{-1}{r}\Big|_{\frac{1}{\sin\theta+\cos\theta}}^1$$

$$= \int_0^{\pi/2} \left(\cos\theta + \sin\theta - 1\right) d\theta = \sin\theta - \cos\theta - \theta\Big|_0^{\pi/2} = 2 - \frac{\pi}{2}.$$

18-1、利用极坐标计算 $\iint_D \frac{dxdy}{1+x^2+y^2}$, $D=\{(x,y)|x^2+y^2\le 1, x\ge 0\}$

解: 依题意, $D: r\le 1, r\cos\theta\ge 0 \Rightarrow \theta\in[\frac{-\pi}{2},\frac{\pi}{2}], r\le 1$

$$\Rightarrow (*)=\int_{-\pi/2}^{\pi/2} d\theta \int_0^1 dr\frac{r}{1+r^2}=\pi, \frac{1}{2}\ln(1+r^2)\Big|_0^1=\frac{\pi}{2}\ln 2$$

18-2、利用极坐标计算 $\iint_D \sqrt{\frac{1-x^2-y^2}{1+x^2+y^2}}\,dxdy$, 其中 D 由坐标轴与圆 $x^2+y^2=1$ 所围成的位于第一象限内的闭区域.

解: 依题意, $D: 0\le\theta\le\frac{\pi}{2}, r\le 1$

$$(*)=\int_0^{\pi/2} d\theta\int_0^1 dr r\sqrt{\frac{1-r^2}{1+r^2}} \qquad 令 t=\sqrt{\frac{1-r^2}{1+r^2}}$$

$$\Rightarrow r^2=\frac{1-t^2}{1+t^2}\Rightarrow 2r\,dr=\frac{-4t\,dt}{(1+t^2)^2}\Rightarrow (*)=\frac{\pi}{2}\int_1^0\frac{-2t^2\,dt}{(1+t^2)^2}$$

$$\int_0^1\frac{dt}{1+t^2}=\frac{t}{1+t^2}\Big|_0^1+\int_0^1\frac{2t^2\,dt}{(1+t^2)^2}\Rightarrow (*)=\frac{\pi}{2}\left(\arctan t\Big|_0^1-\frac{1}{2}\right)=\frac{\pi(\pi-2)}{8}$$